Image-Based Visual Servo Control for Ground Target Tracking

Image-Based Visual Servo

The task of visual servo control is to control the pose of the end-effector relative to the target using visual features extracted from the images. According to the position of the camera,visual servo control is divided into endpoint closed-loop control (hand-eye) and endpoint open-loop control, as shown in Fig. 1.The camera of the drone is attached to its body, so the target tracking control type is endpoint closed-loop control.

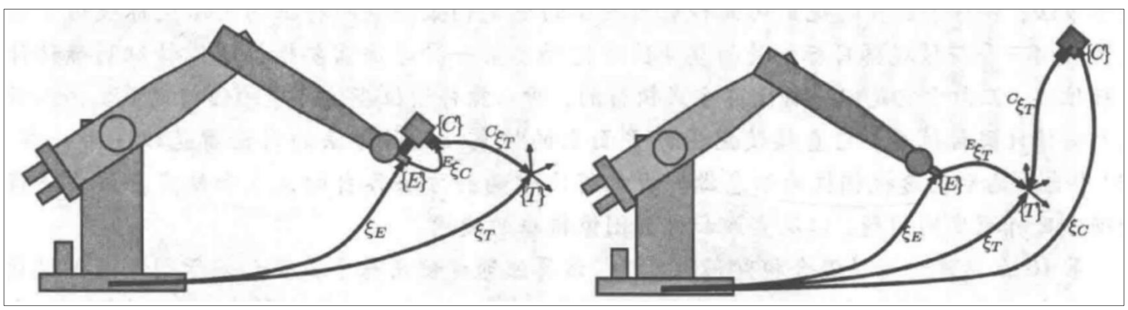

Fig. 1 Endpoint closed-loop control (on the left) and endpoint open-loop control (on the right)

Visual servo control can also be divided into position-based control and image-based control from the difference of controlled parameters. The former needs to estimate the relative pose of the camera and the target, while the latter is based on pixel coordinates of image feature points. In this scheme, the target is regarded as a point on the ground, so it is neither necessary nor impossible to obtain the pose of the target, and it is reasonable to choose image-based visual servo control .

Mapping the point in pixel coordinate system to camera coordinate system is the key of image-based visual servo control , which can be realized by Jacobian matrix transformation shown in equation (1). Pixel velocities are converted to linear velocities and angular velocities in the camera coordinate system, which can be tracked by the UAV and the pan-tilt control system.

$$
\begin{bmatrix} \dot{\bar{u}} \\ \dot{\bar{v}} \end{bmatrix} =
$$

$$
\begin{bmatrix} -\frac{f_x}{z} & 0 & \frac{\bar{u}}{z} & & \frac{\bar{u}\bar{v}}{f_x} & & - \\ 0 & -\frac{f_y}{z} & \frac{\bar{v}}{z} & \frac{f_y^2+\bar{v}^2}{f_y} & -\frac{\bar{u}\bar{v}}{f_y} & -\bar{u} & \end{bmatrix}
$$

(1)

where

$$
\begin{aligned} \bar{u} &= u - u_{center} \\ \bar{v} &= v - v_{center} \end{aligned}
$$

(2)

Using the proportional control, the expected velocities of the pixel are obtained, as shown below:

$$
\begin{aligned} \dot{\bar{u}} &= -k\bar{u} \\ \dot{\bar{v}} &= -k\bar{v} \end{aligned}
$$

(3)

There are six unknowns for linear velocity and angular velocity in equation (1), and we need six equations to solve them.First, a target point pixel provides two equations.In addition, the cradle head remains stable while the UAV is flying, and the camera attitude remains the same ($\omega_x = 0, \omega_y = 0, \omega_z = 0$), so three more equations can be provided.Finally, the drone flies at constant altitude ($v_z = 0$), Normally, when the UAV flies, the camera faces forward with a certain pitch angle. Therefore, when the UAV flies at a certain altitude, the horizontal flight speed of the UAV and the linear speed of the camera are shown in Formula 4.

$$\begin{gathered} T_x = -v_y \\ T_y = -\sin\theta v_x \\ T_z = \cos\theta v_x \end{gathered} \tag{4}$$

Assuming that the target has no height, the depthcan be approximated by Eq. 5.

$$z = \frac{h}{\sin\theta} \tag{5}$$

where h is the flying height of the UAV.

In summary, the control equation of image-based visual servo control is shown in Equation 6.

$$
\begin{aligned}
v_x &= \frac{\dot{\bar{v}} z}{\bar{v}\cos\theta + f_y \sin\theta} \\
v_y &= \frac{z\dot{\bar{u}} - \bar{u}\cos\theta v_x}{f_x}
\end{aligned} \tag{6}
$$

Scheme process

Target detection is performed using YOLOV3, as shown in Fig 2.And the midpoint of the bounding box of a specific class(Human in this scheme) is regarded as the feature point. After the pixel coordinates of the feature point is obtained, the expected UAV speed can be calculated according to Formula (6) and sent to the flight controller to achieve target tracking.

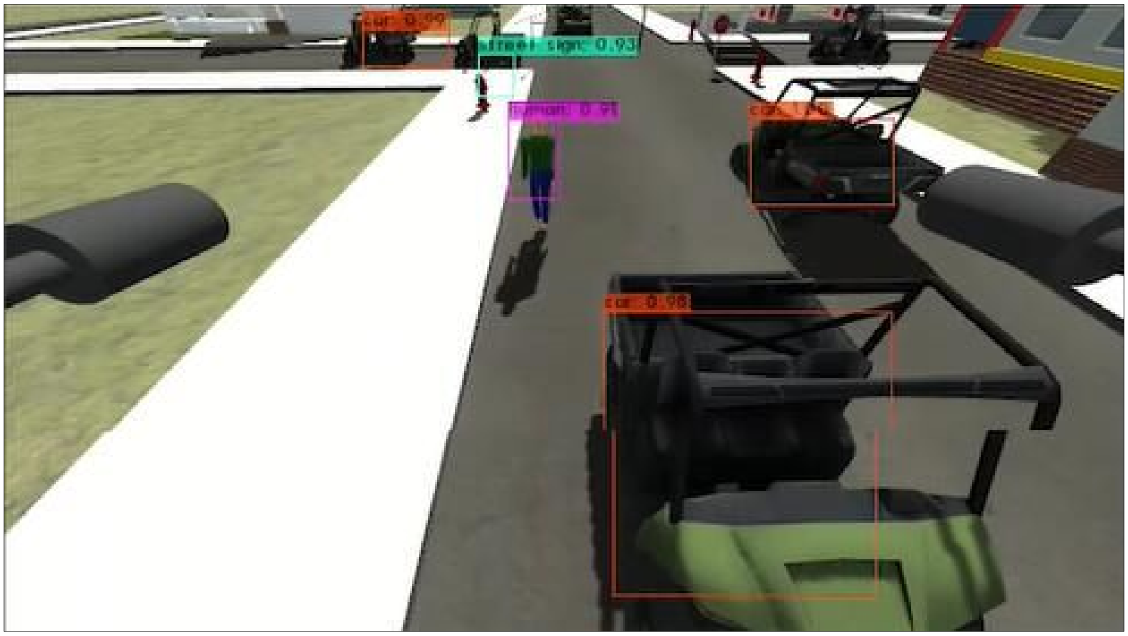

Fig 2. Target detection